

Section 2.6 Linear Transformations

Exercise 2.6.2

Let $T : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ be a linear transformation.

a. Find $T \begin{bmatrix} 1 \\ 3 \\ -2 \\ -3 \end{bmatrix}$ if $T \begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}$ and $T \begin{bmatrix} 0 \\ -1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \\ 1 \end{bmatrix}$.

b. Find $T \begin{bmatrix} 5 \\ -1 \\ 2 \\ -4 \end{bmatrix}$ if $T \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \\ -3 \end{bmatrix}$ and $T \begin{bmatrix} -1 \\ 1 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$.

Exercise 2.6.3

In each case assume that the transformation T is linear, and use Theorem 2.6.2 to obtain the matrix A of T .

b. $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is given by $T(\mathbf{x}) = -\mathbf{x}$ for each $\mathbf{x}$ in $\mathbb{R}^2$.

c. $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is clockwise rotation through $\frac{\pi}{4}$.

Exercise 2.6.4

In each case use Theorem 2.6.2 to obtain the matrix A of the transformation T . You may assume that T is linear in each case.

- b. $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is reflection in the $y - z$ plane.

Exercise 2.6.7

In each case show that $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is not a linear transformation.

a. $T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} y \\ 0 \end{bmatrix}$

b. $T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ y^2 \end{bmatrix}$

Exercise 2.6.8

In each case show that T is either reflection in a line or rotation through an angle, and find the line or angle.

b. $T \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} x + y \\ -x + y \end{bmatrix}$

Section 3.1 The Cofactor Expansion

Exercise 3.1.1

Compute the determinants of the following matrices.

e. $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$

f. $\begin{bmatrix} 2 & 0 & -3 \\ 1 & 2 & 5 \\ 0 & 3 & 0 \end{bmatrix}$

$$\mathbf{h.} \quad \begin{bmatrix} 0 & a & b \\ a & 0 & c \\ b & c & 0 \end{bmatrix}$$

$$\mathbf{k.} \quad \begin{bmatrix} 0 & 1 & -1 & 0 \\ 3 & 0 & 0 & 2 \\ 0 & 1 & 2 & 1 \\ 5 & 0 & 0 & 7 \end{bmatrix}$$

Exercise 3.1.2

Show that $\det A = 0$ if A has a row or column consisting of zeros.

Exercise 3.1.6

Evaluate by cursory inspection:

a. $\det \begin{bmatrix} a & b & c \\ a+1 & b+1 & c+1 \\ a-1 & b-1 & c-1 \end{bmatrix}$

b. $\det \begin{bmatrix} a & b & c \\ a+b & 2b & c+b \\ 2 & 2 & 2 \end{bmatrix}$

Exercise 3.1.7

If $\det \begin{bmatrix} a & b & c \\ p & q & r \\ x & y & z \end{bmatrix} = -1$, compute:

a. $\det \begin{bmatrix} -x & -y & -z \\ 3p+a & 3q+b & 3r+c \\ 2p & 2q & 2r \end{bmatrix}$

b. $\det \begin{bmatrix} -2a & -2b & -2c \\ 2p+x & 2q+y & 2r+z \\ 3x & 3y & 3z \end{bmatrix}$

Exercise 3.1.9

In each case either prove the statement or give an example showing that it is false:

a. $\det(A + B) = \det A + \det B$.

c. If A is 2×2 then $\det(A^T) = \det A$.

e. If A is 2×2 , then $\det(7A) = 49 \det A$.

g. $\det(-A) = -\det A$.

Section 3.2 Determinants and Matrix Inverses

Exercise 3.2.2

Use determinants to find which real values of c make each of the following matrices invertible.

a.
$$\begin{bmatrix} 1 & 0 & 3 \\ 3 & -4 & c \\ 2 & 5 & 8 \end{bmatrix}$$

b.
$$\begin{bmatrix} 0 & c & -c \\ -1 & 2 & 1 \\ c & -c & c \end{bmatrix}$$

c. $\begin{bmatrix} c & 1 & 0 \\ 0 & 2 & c \\ -1 & c & 5 \end{bmatrix}$

d. $\begin{bmatrix} 4 & c & 3 \\ c & 2 & c \\ 5 & c & 4 \end{bmatrix}$

Exercise 3.2.3

Let A , B , and C denote $n \times n$ matrices and assume that $\det A = -1$, $\det B = 2$, and $\det C = 3$. Evaluate:

a. $\det(A^3BC^TB^{-1})$

b. $\det(B^2C^{-1}AB^{-1}C^T)$

Exercise 3.2.5

If A is 3×3 and $\det(2A^{-1}) = -4$ and $\det(A^3(B^{-1})^T) = -4$, find $\det A$ and $\det B$.

Exercise 3.2.6

Let $A = \begin{bmatrix} a & b & c \\ p & q & r \\ u & v & w \end{bmatrix}$ and assume that $\det A = 3$. Compute:

a. $\det(2B^{-1})$ where $B = \begin{bmatrix} 4u & 2a & -p \\ 4v & 2b & -q \\ 4w & 2c & -r \end{bmatrix}$

b. $\det(2C^{-1})$ where $C = \begin{bmatrix} 2p & -a + u & 3u \\ 2q & -b + v & 3v \\ 2r & -c + w & 3w \end{bmatrix}$

Section 3.3 Diagonalization and Eigenvalues

Exercise 3.3.1

In each case find the characteristic polynomial, eigenvalues, eigenvectors, and (if possible) an invertible matrix P such that $P^{-1}AP$ is diagonal.

a. $A = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$

c. $A = \begin{bmatrix} 7 & 0 & -4 \\ 0 & 5 & 0 \\ 5 & 0 & -2 \end{bmatrix}$

e. $A = \begin{bmatrix} 1 & -2 & 3 \\ 2 & 6 & -6 \\ 1 & 2 & -1 \end{bmatrix}$